



# Emulating fundamental analysts: Analytical stage-based multi-agent framework enhanced with expert guidance and Preference-Anchored Likelihood Adjustment

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## ABSTRACT

With the rapid advancement of large language models (LLMs), some studies have explored their potential for predicting stock prices based on financial texts. However, previous research often overlooked the depth of analysis generated by LLMs, resulting in reasoning processes inferior to those of human analysts. In fundamental investing, which requires in-depth company analysis, conclusions from imperfect reasoning lack persuasiveness. In this study, inspired by the analysis process of human analysts, we propose an “Analytical Stage-Based Multi-Agent Framework” to enable LLMs to perform in-depth fundamental analysis. This framework divides the analysis into multiple stages, assigning an LLM agent to each. We enhance each agent’s capabilities for its specific task through expert guidance or fine-tuning, allowing them to collectively emulate the workflow of human analysts. Furthermore, we introduce Preference-Anchored Likelihood Adjustment, a new method for fine-tuning LLMs. This approach addresses the decline in likelihood of generating correct responses that occurs after using existing preference alignment methods. It employs an objective function with two terms: one to increase likelihood and another to preserve aligned preference. We conducted experiments using our framework to analyze company earnings releases. We evaluated the analysis quality based on comprehensiveness and logical soundness, while correctness was assessed by using stock prices as the ground truth to calculate the Matthews correlation coefficient and F1 score. Results demonstrate that even without expert guidance and fine-tuning, our multi-agent framework can enhance LLMs in both analysis quality and correctness. When combined with expert guidance and fine-tuning, the performance is further improved.

## 1. Introduction

The financial market is a crucial area for asset building for individuals and communities, with academic and practical research continually advancing through diverse areas, including asset pricing, portfolio management, derivatives, and et cetera. Investment approaches in the financial market can generally be categorized into fundamental investing and quantitative investing.

Fundamental investing involves assessing the “intrinsic value” of companies or assets based on various factors including company financial statements, business operations, market trends, and macroeconomic environment. This approach places a strong emphasis on a thorough analysis of the intrinsic value of individual stocks (Brealey & Myers, 2003; Valentine, 2011). Quantitative investing makes investment decisions based on vast amounts of quantitative data and leverages mathematical models to process a large number of stocks statistically.

Compared to quantitative investing, fundamental investing demands a deeper analysis of individual companies. Evidence shows that fundamental analysis can achieve investment returns that quantitative analysis alone cannot capture (Bartram & Grinblatt, 2018; Frankel & Lee, 1998; Lee, Myers, & Swaminathan, 1999). Over the past few decades, advancements in technology have significantly enhanced quantitative investing methods and the types of data used (Ferreira, Gandomi, & Cardoso, 2021; Sahu, Mokhadde, & Bokde, 2023). In contrast, fundamental analysis requires extensive examination of unstructured data and broad economic conditions (Valentine, 2011), making it more challenging to apply analytical techniques. As a result, fundamental analysts still spend substantial effort on workflows that have barely changed in decades, which highlighting the significant value in developing technologies to assist human analysts for fundamental analysis.

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In the field of natural language processing (NLP), numerous studies have focused on analyzing stock prices using textual information. For instance, researchers have explored methods that employ Encoder models like BERT to convert text into embeddings, which are then utilized for sentiment analysis or integrated with other data to predict stock prices (Ashtiani & Raahemi, 2023; Ding, Zhang, Liu, & Duan, 2015; Luo, Liao, Li, Cheng, & Yan, 2023). However, the reasoning processes of encoder models lack readability, making it difficult for users to understand how conclusions are reached, which poses challenges for their application in fundamental analysis. LLMs have demonstrated a range of capabilities in tasks like sentiment analysis, mathematical reasoning, and question answering in the financial domain Li et al. (2023), Niszczota and Abbas (2023), Yang, Tang, and Tam (2023), Zhao et al. (2024), and empirical studies have shown that these abilities allow them to present arguments with a certain level of persuasiveness to financial experts (Takayanagi, Suzuki, Kobayashi, Sakaji, & Izumi, 2024; Takayanagi, Takamura, Izumi, & Chen, 2024). Therefore, some researchers have fed financial texts into LLMs and let them predict stock prices (Ding, Li, Wang, & Chen, 2024; Fatouros, Metaxas, Soldatos, & Kyriazis, 2024; Lopez-Lira & Tang, 2023; Zhang et al., 2023). Although LLMs can generate readable reasoning processes, they have not been specifically trained for stock price analysis. Therefore, some studies aim to enhance their predictive capabilities by providing LLMs with instructions. For instance, LLMFactor (Wang, Izumi, & Sakaji, 2024) prompts LLMs to extract profitable factors from news articles before stock price prediction. Other studies focus on enhancing the stock price prediction capabilities through model fine-tuning, as demonstrated by Yu, Chen, and Lu (2023). However, previous studies typically employ only one of the two approaches, each suited to enhance different capabilities:

1. Prompting: This approach is effective for enabling LLMs to efficiently acquire more specialized “how-to” knowledge, which refers to analysis methods that can be expressed in written form. For example, it can guide LLMs on how to calculate various financial metrics using data from financial statements.
2. Fine-tuning: This approach is suitable for enhancing specific skills that are challenging to articulate with clear rules. For example, it can enhance an LLM’s skill in identifying which pieces of information are important within complex and varied data. However, fine-tuning is suitable for tasks with correct answers and requires a sufficient amount of training data.

Human analysts use both of the above-mentioned approaches to enhance their analytical abilities. Prompting is akin to learning about analytical methods through books or other resources. Fine-tuning, is analogous to improving one’s judgment abilities through experience. Therefore, if both approaches can be applied to LLMs in the process of fundamental analysis, it would result in a more comprehensive improvement of their analytical capabilities.

However, just prompting and fine-tuning are not sufficient to enable LLMs to achieve in-depth fundamental analysis. Previous studies have often focused more on the accuracy of stock price prediction, without paying sufficient attention to the depth of analysis during the reasoning process. As a result, the reasoning processes of stock price prediction demonstrated by LLMs are much inferior compared to that of human analysts. What is more important is the process of analysis. In practice, fundamental analysts typically derive conclusions through a series of analyses from various perspectives. For example, suppose a company discloses that “this quarter’s profit is \$1 billion, a 5% increase year-over-year”. To determine whether this information is positive or negative, an analyst would first need to calculate the progress rate of this quarter’s profit relative to the full year profit forecast and then compare this with the progress rate in the same quarter of previous years to assess whether the profit is high or low. Further analysis would involve determining whether the profit growth is due to business expansion or accounting adjustments to judge whether it is sustainable. Only through such thorough, step-by-step analyses, can one assess

the positivity or negativity of the information from a fundamental perspective. We believe that enabling LLMs to perform such analyses in a process similar to human will make their analyses more valuable.

As discussed above, when performing fundamental analyses, human analysts follow multiple steps to arrive at a final conclusion, employing different skills along the way. Therefore, to let LLMs perform in-depth fundamental analysis, we propose an Analytical Stage-Based Multi-Agent Framework (Fig. 1). This framework breaks down the analysis process into three stages, and assigns an LLM agent to each stage. Stage 1: the Financial Analyzer agent performs a quantitative analysis of the company’s financial data. Stage 2: the Business Analyzer performs a comprehensive analysis of business performance by combining the results from Stage 1 and qualitative information from the company’s releases; Stage 3: the Conclusion Deriver identifies key points from the analysis in Stage 2 and deduce a conclusion about the company’s future prospect. Since the analysis methods for stage 1 and 2 can be articulated in words, we prompt expert guidance about analysis methods to enhance the analytical capability of Financial Analyzer and Business Analyzer. For Stage 3, as the skills required are difficult to express through a set of rules in words, fine-tuning is employed to enable Conclusion Deriver to learn from real-world cases. Together, in sequence, these agents emulate the complete analytical workflow of a human analyst.

Here, to fine-tune the Conclusion Deriver, we propose an automatic way to construct the training data and a new method for fine-tuning. First, regarding training data, two sets are prepared: one where the analysis concludes with correct future prospects, and another where it concludes with incorrect future prospects. We provide an LLM with either correct or incorrect future prospects and instruct it to perform analysis based on the analysis generated by the Business Analyzer in such a way that it concludes with the provided prospects. The correct or ground truth prospects are defined based on stock price trends. In this manner, we utilize the LLM to automatically construct the training data. Second, for fine-tuning, we propose a new method: Preference-Anchored Likelihood Adjustment. After preference alignment for LLMs with existing methods, the likelihood of correct responses may decline compared to the model before fine-tuning. By introducing a novel objective function, our method adjusts for this decline in likelihood while maintaining the fine-tuned preference, thereby enhancing the LLM’s analysis capabilities for this task.

Note that we do not think that LLMs can perform all analysis tasks at the same level as human professionals. This is due to the limitations of LLMs capabilities and the challenge of accessing various types of information that human gather, not just from company releases or news but also through conversations with market participants and companies. Therefore, the goal of this study is to integrate AI into the fundamental analysis workflow to reduce human effort while providing valuable references for decision-making. As discussed in Qiu and Jin (2024), an integration of AI with human expertise can substantially enhance the efficiency of domain experts in accomplishing their objectives.

Our main contributions are as follows:

1. In the context of limited attention to the use of LLMs for in-depth fundamental analysis, we propose the Analytical Stage-Based Multi-Agent Framework to enable such in-depth analysis. Our framework breaks down the analysis process into multiple stages. For each stage, we assign an agent with capability enhanced for its specific task.
2. To fine-tune the Conclusion Deriver agent, we propose an automatic way to construct the training data and a new method for fine-tuning called Preference-Anchored Likelihood Adjustment to adjust the model likelihood of generating correct responses.
3. We validate the effectiveness of our framework through experiments on listed companies. Even without expert guidance and fine-tuning, our proposed multi-agent framework can enhance LLMs in both the quality of analysis and the correctness of conclusions. When enhancing the agents with expert guidance and our proposed fine-tuning, the performance is further improved.

The remainder of this paper is organized as follows: Section 2 discusses related works. Section 3 explains our proposed framework. Next, Section 4 discusses experiment settings and the results. Finally, Section 5 concludes this paper.

## 2. Related works

### 2.1. Language agents

LLMs have demonstrated human-level performance on various benchmarks (Achiam et al., 2023; Dubey et al., 2024), spurring extensive studies on LLM agents (Wang, Ma, et al., 2024; Xi et al., 2023). An LLM agent is a system designed to autonomously execute tasks by leveraging LLMs. By integrating with databases and external tools, these agents can augment LLMs capabilities. For example, combining the reasoning capabilities of LLMs with retrieval of external knowledge (Lewis et al., 2020) enables the execution of knowledge-intensive tasks (Gao et al., 2022; Press et al., 2022; Trivedi, Balasubramanian, Khot, & Sabharwal, 2022; Yao et al., 2022). CoALA (Sumers, Yao, Narasimhan, & Griffiths, 2024) organizes various proposed language agents within a framework inspired by cognitive architectures. Cognitive architecture refers to theoretical models that aim to replicate the structures and processes of human cognition. Notable examples include Soar (Laird, 2022) and ACT-R (Ritter, Tehranchi, & Oury, 2019). Compared to single-agent systems, multi-agent systems utilize multiple agents that collaborate to complete tasks. Each agent is specialized in a specific task, enabling the system as a whole to possess greater capabilities than a single-agent system (Guo et al., 2024).

### 2.2. Stock price analysis using textual information

Numerous studies have utilized textual information for stock price analysis. Research utilizing Encoder models like BERT typically involves converting textual information into embeddings, which are then fed into deep learning frameworks to predict stock prices. Some studies extract company-related events from news articles for stock price prediction (Ding et al., 2015), incorporate inter-company relationships (Cheng, Yang, Xiang, & Liu, 2022; Xu et al., 2021), and consider the temporal dynamics of news (Sawhney, Wadhwa, Agarwal, & Shah, 2021) as well as related stock price data (Luo et al., 2023). However, the inference process in these methods is not human-interpretable, making them less compatible with fundamental investing.

Studies utilizing LLMs have reported higher performance compared to Encoder models. Some studies have involved feeding company news into LLMs to determine sentiment (Kirtac & Germano, 2024; Lopez-Lira & Tang, 2023; Yang, Liu, & Wang, 2023; Zhang et al., 2023). In LLMFactor (Wang, Izumi, & Sakaji, 2024), LLMs were instructed to extract profitable factors from news before stock price prediction. MarketSenseAI (Fatouros et al., 2024) not only use company news but also releases and macroeconomic information, summarizing them with LLMs before making stock predictions. Other studies have enhanced LLMs' reasoning capabilities through instruction-tuning (Yu et al., 2023) or reinforcement learning (Koa, Ma, Ng, & Chua, 2024). While some studies have used prompts to assist LLMs in stock analysis or fine-tuned LLM models to enhance their stock analysis capabilities, the focus has predominantly been on improving the accuracy of stock price predictions, rather than the depth of fundamental analysis.

This study aims to enable LLMs to engage in in-depth analytical processes similar to those of human analysts, producing analyses that can be informative for fundamental investing.

### 2.3. Preference alignment for LLMs

Preference alignment is the process of fine-tuning LLMs to produce outputs that adhere to specific preferences, such as following intended instructions, generating higher-quality responses, and ensuring harmlessness. Reinforcement learning from human feedback (RLHF) is a typical approach (Ouyang et al., 2022; Schulman, Wolski, Dhariwal, Radford, & Klimov, 2017; Stiennon et al., 2022). RLHF involves first training a reward model based on human feedback and then using this reward model to guide reinforcement learning. DPO (Direct Preference Optimization) (Rafailov, Sharma, Mitchell, Ermon, Manning, & Finn, 2024) pointed out that RLHF is complex and often unstable, and proposed a cross-entropy objective to directly align LLMs with human preferences without explicit reward modeling. Following this, various improved methods based on DPO have been developed (Azar, Rowland, Piot, Guo, Calandriello, Valko, & Munos, 2023; Chen et al., 2024; Chowdhury, Kini, & Natarajan, 2024; D'Oosterlinck, Xu, Devellder, Demeester, Singh, Potts, Kiela, & Mehri, 2024; Ji et al., 2024; Jung, Han, Nam, & On, 2024; Liu et al., 2024; Melnyk et al., 2024; Pang, Yuan, Cho, He, Sukhbaatar, & Weston, 2024; Wu, Sun, Yuan, Ji, Yang, & Gu, 2024). One such method, APO (Anchored Preference Optimization) (D'Oosterlinck et al., 2024), introduced a controllable and more stable alignment objective.

Our study aims to enhance the capability of LLMs for company analysis. After fine-tuning LLMs with existing alignment methods, we introduce a new objective function for further fine-tuning LLMs: Preference-Anchored Likelihood Adjustment.

## 3. Methodology

In this section, we explain the three agents in our proposed Analytical Stage-Based Multi-Agent Framework (Fig. 1), along with our fine-tuning method, "Preference-Anchored Likelihood Adjustment", and the memory base that record all information needed for the agents.

### 3.1. Agents description

When performing analyses, human fundamental analysts follow multiple steps to arrive at a final conclusion. While the procedures may vary somewhat among analysts, in this study we follow a common practice and divided the analysis process into the following three stages: (1) quantitative analysis of financial data; (2) a comprehensive analysis based on analysis performed in stage 1 and qualitative information from earnings reports; and (3) identifying key points from analysis performed in stage 2 and reaching a final conclusion for future outlook. We propose a multi-agent framework to emulate such process. Capability of each LLM agent is enhanced with expert guidance or fine-tuning. Expert guidance refers to text descriptions of analysis methods used by human analysts (Appendix Fig. 4). Since we observed that LLMs have yet to exhibit analytical capabilities comparable to those of professional human analysts, we explicitly add these expert guidance to their prompts to refine their analytical approach. The description of each agent is as below.

**Financial Analyzer:** To evaluate the financial performance for a given fiscal period, it is important to calculate various financial ratios or compare financial data with historical records. This includes but is not limited to calculating growth rates of revenue and profits, progress rate towards company guidance, and changes in dividend. Such analysis provides a deeper understanding of a company's financial performance. Human analysts typically acquire these analytical methodologies from books or other experienced analysts. Similarly, we summarize these analytical approaches into a paragraph which we refer to as expert guidance (Appendix Fig. 4(a)), and provide it with a prompt to the Financial Analyzer. The Financial Analyzer then uses the financial data from the current fiscal quarter, the preceding quarter, and the corresponding quarter from the prior year to perform the above-mentioned

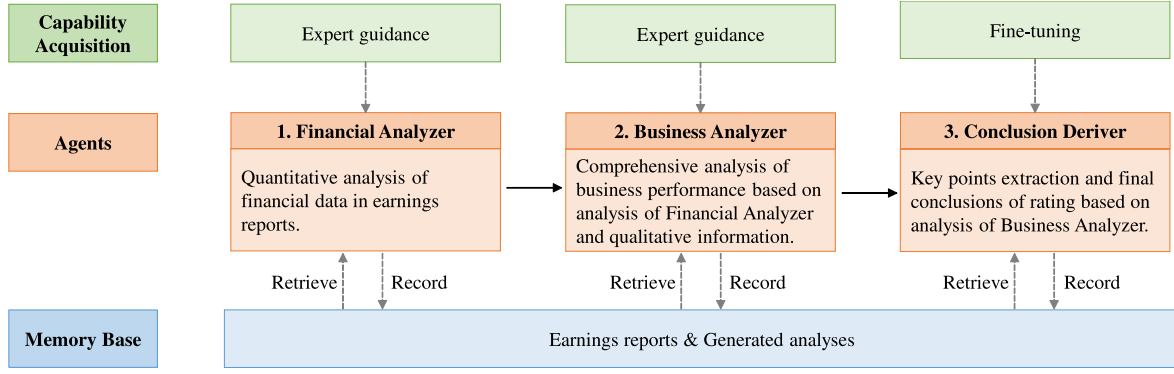


Fig. 1. Our proposed Analytical Stage-Based Multi-Agent Framework.

analysis. The prompt for the Financial Analyzer is shown in Appendix Fig. 5(a). Subsequently, this quantitative analysis will be passed to the agent in the next stage through prompt, and serve as a foundation for more in-depth analysis.

**Business Analyzer:** The Business Analyzer plays a crucial role in providing a comprehensive understanding of a company's overall fundamentals. It utilizes qualitative information from earnings reports, combined with financial analysis from the Financial Analyzer. We prompt the agent to firstly generate an analysis outline based on the information provided. We enhance this process by add expert guidance (Appendix Fig. 4(b)) to the prompt to ensure that essential analysis points are covered, such as analyzing the reasons behind earnings changes or analyzing the shareholder return policy. We then let it conduct analysis based on the outline. The prompt for the Business Analyzer is shown in Appendix Fig. 5(b). The generated comprehensive analysis will be passed to the next agent through prompt,

**Conclusion Deriver:** From the comprehensive analysis generated by the Business Analyzer, we let Conclusion Deriver identify key points and then assign one of the following ratings to the company: "Positive: performance is expected to improve" or "Negative: performance is expected to decline". Identifying key points can help the agent to better deduce the rating. Since the importance of the key points varies depending on the situation, it is difficult to define a set of rules in words to instruct the Conclusion Deriver. Therefore, we use fine-tuning to enhance the Conclusion Deriver's capability for this particular task. Section 3.2 illustrates the details of fine-tuning. The prompts for Conclusion Deriver is shown in Appendix Fig. 5(c).

### 3.2. Preference-anchored likelihood adjustment

Just as humans can learn from both successful and unsuccessful experiences, we leverage both correct and incorrect analyses as training data. These contrastive data pairs can be seen as preference pairs, and we fine-tune an LLM to align with the preference for generating correct analysis.

First, we create correct analyses (winning responses) and incorrect analyses (losing responses) in the stage of the Conclusion Deriver. In this paper, we focus on company analysis conducted after earnings announcement. If the conclusion of the company analysis is that the company's performance will improve (or decline), and the excess return of the stock price after the earnings announcement is positive (or negative), it is considered winning response. Otherwise, it is considered losing response. This approach assumes that the short-term stock price after the earnings announcement accurately reflects the company's prospect. The excess return after day  $t$  is defined below:

$$ExR_t = S_t/S_0 - M_t/M_0 \quad (1)$$

$S_0$  is the closing price of the target company on the day of the earnings announcement,  $S_t$  is the closing price after  $t$  days, and  $M_0$  and  $M_t$

are the respective closing prices of the market index. Excess return is used to remove the effect of overall market fluctuations. To eliminate noise, the training and validation data are limited to samples where the direction of excess return on  $t = 1$  and  $t = 5$  are the same. To generate a winning response, we provided GPT-4o mini with the direction of  $ExR_1$  and the comprehensive company analysis generated by Business Analyzer and had it derive a conclusion that matches this direction. To generate a losing response, we provided the opposite direction and had GPT-4o mini derive conclusion accordingly. Comparing to SEP (Koa et al., 2024) that uses an iterative self-reflective (Shinn, Cassano, Berman, Gopinath, Narasimhan, & Yao, 2023) approach to have an LLM generate the data, our method is more straightforward and easy to implement. Using the above procedure, we create tuples  $(x, y_w, y_l)$  of a comprehensive analysis  $x$ , a winning response  $y_w$ , and a losing response  $y_l$ .

Next, we fine-tune an LLM in three steps: 1. SFT (supervised fine-tuning), to learn what the winning responses should be; 2. APO (Anchored Preference Optimization) (D'Oosterlinck et al., 2024), to learn the preference for winning responses over losing responses; and 3. our proposed PALA (Preference-Anchored Likelihood Adjustment), to increase the likelihood of the model to generate a winning response.

We use the winning output  $(x, y_w)$  as the training data for SFT. The fine-tuned model is then used as the base model for fine-tuning with APO. Following DPO (Rafailov et al., 2024), APO defines the likelihood change of responses during training as the reward:

$$r_\theta(x, y) = \beta \log \frac{\pi_{apo}(y|x)}{\pi_{sft}(y|x)} \quad (2)$$

Here,  $\pi_{sft}$  is the reference model fine-tuned with SFT,  $\pi_{apo}$  is the model under fine-tuning with APO,  $\beta$  is a parameter controlling the deviation of  $\pi_{apo}$  from the  $\pi_{sft}$ , and  $\pi(y|x)$  is the likelihood that a model generates  $y$  when given  $x$  as a prompt. Then, APO applies the following objective to increase the reward for winning responses and decrease the reward for losing responses, i.e. increase the likelihood of generating winning responses while decrease the likelihood of generating losing responses:

$$\mathcal{L}_{APO}(x, y_w, y_l; \theta) = -\sigma(r_\theta(x, y_w)) + \sigma(r_\theta(x, y_l)) \quad (3)$$

where  $\sigma$  is a sigmoid function,  $y_w$  is the winning response and  $y_l$  is the losing response. However, with this objective, there is a possibility that while the losing reward decreases and the margin between winning and losing reward increases, the winning reward undesirably decreases. We observed this phenomena in validation and test dataset in our experiment. This means that although the model  $\pi_{apo}$  have learned the difference between winning and losing responses, its likelihood of generating winning responses become lower than that of the reference model  $\pi_{sft}$ .

Therefore, to adjust the decline in likelihood while maintaining the margin between winning and losing responses, we designed a objective function described below to further fine-tune the model:

$$\mathcal{L}_{PALA}(x, y_w, y_l; \theta') = -\sigma(r_{\theta'}(x, y_w)) - \text{Anc}(r_{\theta'}(x, y_l)) \quad (4)$$

where,

$$rm_{\theta'}(x, y) = r_{\theta'}(x, y_w) - r_{\theta'}(x, y_l) \quad (5)$$

$$r_{\theta'}(x, y) = \beta \log \frac{\pi_{pala}(y|x)}{\pi_{apo}(y|x)} \quad (6)$$

$$\text{Anc}(x) = \begin{cases} \sigma(x), & \text{if } x \geq 0 \\ a * \exp(k * x) + b, & \text{if } x < 0 \end{cases} \quad (7)$$

$\pi_{apo}$  is the model fine-tuned with APO,  $\pi_{pala}$  is the model under fine-tuning,  $r_{\theta'}(x, y)$  is the reward of  $\pi_{pala}$  against  $\pi_{apo}$ ,  $rm_{\theta'}(x, y)$  is the reward margin between winning and losing responses, and  $a, k, b$  are set such that  $\text{Anc}(x)$  remains continuously differentiable at 0.

The first term of the objective function serves to increase the reward of winning responses, thus adjust the likelihood decline during APO fine-tuning. The second term is a function of the reward margin. Substituting Eq. (6) into (5), the reward margin can be written as:

$$rm_{\theta'}(x, y) = \beta \log \frac{\pi_{pala}(y_w|x)}{\pi_{apo}(y_w|x)} - \beta \log \frac{\pi_{pala}(y_l|x)}{\pi_{apo}(y_l|x)} \quad (8)$$

Similarly, the reward margin for  $\pi_{apo}$  against  $\pi_{sft}$  can be written as:

$$rm_{\theta}(x, y) = \beta \log \frac{\pi_{apo}(y_w|x)}{\pi_{sft}(y_w|x)} - \beta \log \frac{\pi_{apo}(y_l|x)}{\pi_{sft}(y_l|x)} \quad (9)$$

By summing up Eq. (8) and (9) for each side, we can get

$$rm_{\theta}(x, y) + rm_{\theta'}(x, y) = \beta \log \frac{\pi_{pala}(y_w|x)}{\pi_{sft}(y_w|x)} - \beta \log \frac{\pi_{pala}(y_l|x)}{\pi_{sft}(y_l|x)} \quad (10)$$

The RHS of Eq. (10) is the reward margin of  $\pi_{pala}$  against  $\pi_{sft}$ . If  $rm_{\theta'}(x, y) < 0$ , the reward margin of  $\pi_{pala}$  against  $\pi_{sft}$  is less than that of  $\pi_{apo}$  against  $\pi_{sft}$ , which should be avoided.  $\text{Anc}(x)$  takes a form that if  $rm_{\theta'}(x, y)$  moves under 0, it rapidly decreases if parameters  $a$  and  $k$  are negative numbers and are appropriately set, leading to the increase in the objective function. Thus the second term in Eq. (4) aims to prevent the preference learned in APO from diminishing. Further discussions about the objective function and how reward changes during training can be found in Appendix A.2.

Thus, our proposed PALA fine-tunes the model to increase the winning reward while preserving the preference learned in APO.

### 3.3. Memory base

Memory base records all information in the form of a graph. Each node in the graph consists of a label and content. The content records the main information, while the label records the attributes of the information, such as the company name, fiscal period, and type of information (financial data, qualitative information, generated analyses). The information needed by each agent for analysis is extracted according to the rules described in Appendix Fig. 8. Initially, the financial data and qualitative information for each company and each fiscal period are stored in their respective nodes. The analysis results from each agent are added as new nodes to the graph.

## 4. Experiments

### 4.1. Data

To conduct fundamental analysis, abundant company information is essential. Therefore, in this paper, we analyze companies using their earnings reports which provide detailed disclosures. The earnings reports are originally downloaded from company websites, and stock price data are obtained from Bloomberg. We convert the earnings reports from PDF to text, extracting key financial data for input into the Financial Analyzer and other qualitative information for input into the Business Analyzer. The financial data and qualitative information used are listed in Appendix Fig. 7. The experiments focus on the earnings reports of Japanese companies announced between January 2022 and May 2024. We randomly selected 3000 samples for training, 300 for validation, and 600 for test.

### 4.2. Baselines

To test the effectiveness of our proposed multi-agent framework, we compare the following four settings:

- **Setting 1. Single-agent without expert guidance and fine-tuning:** This presents the native capability of LLMs in fundamental analysis. We provide an LLM with financial data and qualitative information in earnings reports and prompt the LLM to analyze the company performance, extract positive/negative key points and then deduce the rating. We append the instruction “Think step-by-step” to the end of the prompt, as demonstrated in Yu et al. (2023) that this CoT approach can improve the performance. The analysis is performed in a single response. Experiments were conducted using two LLMs capable of handling long texts and supporting the Japanese language: GPT-4o mini (Achiam et al., 2023) and Qwen2-7B (Yang, Yang, Hui, Zheng, Yu, et al., 2024).
- **Setting 2. Multi-agent without expert guidance and fine-tuning:** This tests the impact of applying three agents according to our proposed way of breaking down the analysis process. We utilize the same three agents as described in Section 3.1, but do not apply expert guidance and fine-tuning.
- **Setting 3. Multi-agent with expert guidance:** This tests the impact of enhancing the capabilities of agents with expert guidance. We apply expert guidance for Financial Analyzer and Business Analyzer. Experiments were conducted using GPT-4o mini and Qwen2-7B. For more comparison, we test LLMs specifically trained for the Japanese language: Llama-3-ELYZA-8B (Hirakawa, Horie, Nakamura, Oba, Pas-saglia, & Sasaki, 2024) and Llama-3-youko-8b (Sawada et al., 2024). Since the lengths of earnings reports often exceeds the max token limit of these two LLMs, we utilize them only with the Conclusion Deriver agent. We apply the analysis generated by Business Analyzer powered with GPT-4o mini as the input for these two LLMs.
- **Setting 4. Multi-agent with expert guidance and our proposed fine-tuning:** This is the complete implementation of our proposed framework. This further tests the impact of applying our proposed fine-tuning method. We use the analysis generated by Business Analyzer powered with GPT-4o mini as the input, and employ the fine-tuned LLM to drive Conclusion Deriver. The training specifications for fine-tuning are stated below.

### 4.3. Training specifications

We train the LLM based on TRL library.<sup>1</sup> Qwen2-7B is used as the base model. SFT, APO, and our proposed PALA were each conducted for 2 epochs, with learning rates set to 3e-4, 1e-5, and 5e-6 respectively, using 4-bit QLoRA (Dettmers, Pagnoni, Holtzman, & Zettlemoyer, 2023).  $\beta$  in reward for APO and PALA is set to 0.5. Parameters for  $\mathcal{L}_{PALA}$  were set as follows:  $a = -0.125$ ,  $k = -2$ , and  $b = 0.625$ . The final SFT model was chosen based on the minimum validation loss, the final APO and PALA models were selected based on the highest reward margin checkpoint of validation data. Inference on the test data was performed using greedy decoding with a temperature of 0.

### 4.4. Evaluation metrics

The purpose of this study is to generate in-depth analyses that can be useful for fundamental investing. Therefore, we evaluate the analyses generated by LLMs from two perspectives.

<sup>1</sup> <https://huggingface.co/docs/trl/index>

**Overall Analysis**

- 1. Comprehensiveness: Whether all aspects of performance are covered.
- 2. Detail: Whether detailed information is presented about each aspect.
- 3. Clarity: Whether the report's structure is logical and clearly organized.

**Final Rating**

- 4. Comprehensiveness: Whether various factors are considered for rating.
- 5. Logical soundness: Whether the rating is derived logically.

Fig. 2. Assessment criteria for the quality of the analysis.

**Table 1**

Correctness of conclusions across different settings. Bold values represent the highest performance across all settings.

|                                                           |                  | Trading Day 0-1 |              | Trading Day 0-5 |              | Trading Day 0-10 |              | Trading Day 0-20 |              |
|-----------------------------------------------------------|------------------|-----------------|--------------|-----------------|--------------|------------------|--------------|------------------|--------------|
|                                                           |                  | MCC             | Macro F1     | MCC             | Macro F1     | MCC              | Macro F1     | MCC              | Macro F1     |
| Setting 1: Single-agent w/o guidance and finetuning       | GPT-4o mini      | 0.034           | 0.502        | 0.028           | 0.502        | 0.029            | 0.502        | -0.006           | 0.487        |
|                                                           | Qwen2-7B         | 0.042           | 0.521        | 0.034           | 0.517        | 0.032            | 0.516        | 0.029            | 0.515        |
| Setting 2: Multi-agent w/o guidance and finetuning        | GPT-4o mini      | 0.092           | 0.541        | 0.064           | 0.529        | 0.121            | 0.557        | 0.071            | 0.533        |
|                                                           | Qwen2-7B         | 0.093           | 0.533        | 0.066           | 0.517        | 0.068            | 0.518        | 0.042            | 0.503        |
| Setting 3: Multi-agent with guidance                      | GPT-4o mini      | 0.156           | 0.570        | 0.154           | 0.566        | 0.163            | 0.571        | 0.054            | 0.515        |
|                                                           | Qwen2-7B         | 0.109           | 0.549        | 0.112           | 0.548        | 0.121            | 0.553        | 0.079            | 0.530        |
| (Based on analysis by Business Analyzer with GPT-4o mini) | Llama-3-ELYZA-8b | 0.042           | 0.515        | 0.056           | 0.524        | 0.056            | 0.523        | -0.018           | 0.488        |
|                                                           | Llama-3-youko-8B | 0.079           | 0.527        | 0.099           | 0.533        | 0.101            | 0.535        | -0.000           | 0.483        |
| Setting 4: Multi-agent with guidance & finetuning         |                  | <b>0.180</b>    | <b>0.583</b> | <b>0.184</b>    | <b>0.582</b> | <b>0.173</b>     | <b>0.577</b> | <b>0.111</b>     | <b>0.544</b> |

**4.4.1. Correctness of conclusions**

We use the direction of the excess return  $ExR_t$  after the earnings announcement as the gold answer for the positive/negative rating deduced by the Conclusion Deriver. The time periods  $t$  correspond to 1, 5, 10, and 20 trading days, which roughly equate to 1 day, 1 week, 2 weeks, and 1 month in calendar days. We use the Matthews Correlation Coefficient (MCC) and macro F1 score as evaluation metrics, as they account for data imbalance, given that positive and negative ratings in company analysis are equally important.

**4.4.2. Quality of analysis**

A persuasive analysis derives conclusions by adequately considering various aspects of companies and using logical reasoning. Therefore, a qualitative evaluation is conducted based on five criteria shown in Fig. 2: comprehensiveness, detail and clarity of the overall analysis, as well as the comprehensiveness and logical soundness of the rationale for the final conclusion. We use GPT-4o mini to do the evaluation. Since assigning an absolute score to each single analysis could not catch the difference among analyses, we let the LLM compare two analyses and determine which one is better for each criteria.

Specifically, when comparing analyses generated by two models, let the number of data samples be  $N$ . For criterion  $i$ , if Model 1 performs better on  $n_i^1$  samples and Model 2 performs better on  $n_i^2$  samples, the relative evaluation of Model 1 versus Model 2 for criterion  $i$  is defined as  $(n_i^1 - n_i^2)/N$ . Note that GPT-4o mini has a tendency to rate the analysis that appears first more favorably, so the evaluation is conducted twice with reversed order, and the average score is taken.

**4.5. Results**

Table 1 presents the correctness of the conclusions, while displays the quality of the analyses.

**Setting 2 vs. Setting 1:** Without using expert guidance or fine-tuning, our proposed multi-agent approach shows better performance compared to the single-agent approach in all data points except one, over a period of up to 20 days for both GPT-4o mini and Qwen2-7B (Table 1). The only exception is the macro F1 score for trading days 0–20 using Qwen2-7B, where both approaches exhibit similarly low values. Regarding the analysis quality (Table 2(a)), multi-agent

approach outperforms the single-agent approach in all criteria. These results indicate that breaking down the analysis process into multiple stages, similar to the approach used by human analysts, and assigning an agent to each stage, can lead to more comprehensive, detailed, and logical analyses. And the higher quality analyses improves the accuracy of conclusions.

**Setting 3 vs. Setting 2:** With the incorporation of expert guidance, both the overall correctness of conclusions (Table 1) and the quality of analyses (b)) improved for both GPT-4o mini and Qwen2-7B. Thus, for analysis methods that can be articulated in words, prompting is an effective way to enhance the capabilities of agents.

**Setting 4 vs. Setting 3:** With the incorporation of our proposed fine-tuning method, PALA, the correctness of conclusions improved across all data points over a period of up to 20 days (Table 1). Moreover, the metrics consistently maintained a certain level of performance, indicating that our proposed framework can analyze the fundamental aspects of a company rather than short-term, unsustainable factors. Note that while the fine-tuning is based on the Qwen2-7B, which has a much smaller size than GPT-4o mini, the performance is better. Also, the fine-tuned model exceeds the performance of Llama-3-ELYZA-8b and Llama-3-youko-8B, which are specifically trained for the Japanese language. This underscores the importance of domain specific fine-tuning. Regarding the analysis quality, since fine-tuning is applied to the Conclusion Deriver, which only deduces the conclusions, we compare only criteria 4–5 that evaluate these conclusions. The incorporation of fine-tuning improves the analysis quality compared to other models (c)). An example of analysis generated by our multi-agent framework can be found in Appendix Fig. 6. We can confirm that various aspects of the company are analyzed and then positive and negative factors are extracted to deduce the final conclusion. We also conducted supplementary experiments for the PALA fine-tuning using other set of train, validation and test data selected from earnings reports of Japanese companies with different random seeds, and got the consistent results.

Note that, since many Japanese companies announce their earnings after trading hours during the experiment period between January 2022 and May 2024, the conclusions from the analysis are not equivalent to trading signals. Because we cannot use the conclusions to trade at the close prices before earnings release. Trading strategies

**Table 2**

Quality of analysis across different settings. Numbers indicates the percentage of samples that the first setting is better than the second setting.

| (a) Setting 2 (multi-agent w/o guidance and fine-tuning) vs.<br>Setting 1 (single-agent w/o guidance and fine-tuning) |     |                  |          |     |     |     |     |   |
|-----------------------------------------------------------------------------------------------------------------------|-----|------------------|----------|-----|-----|-----|-----|---|
| Setting 2                                                                                                             |     | Setting 1        | Criteria | 1   | 2   | 3   | 4   | 5 |
| GPT-4o mini                                                                                                           |     |                  |          |     |     |     |     |   |
| Multi-agent                                                                                                           | vs. | Single-agent     | 100%     | 76% | 99% | 46% | 47% |   |
| Qwen2-7B                                                                                                              |     |                  |          |     |     |     |     |   |
| Multi-agent                                                                                                           | vs. | Single-agent     | 99%      | 76% | 58% | 97% | 83% |   |
|                                                                                                                       |     |                  |          |     |     |     |     |   |
| (b) Setting 3 (multi-agent with guidance) vs.<br>Setting 2 (multi-agent w/o guidance and fine-tuning)                 |     |                  |          |     |     |     |     |   |
| Setting 3                                                                                                             |     | Setting 2        | Criteria | 1   | 2   | 3   | 4   | 5 |
| GPT-4o mini                                                                                                           | vs. | GPT-4o mini      | 67%      | 88% | 62% | 39% | 61% |   |
| Qwen2-7B                                                                                                              | vs. | Qwen2-7B         | 48%      | 40% | 0%  | 30% | 1%  |   |
|                                                                                                                       |     |                  |          |     |     |     |     |   |
| (c) Setting 4 (multi-agent with guidance and fine-tuning) vs.<br>Setting 3 (multi-agent with guidance)                |     |                  |          |     |     |     |     |   |
| Setting 4                                                                                                             |     | Setting 3        | Criteria | 4   | 5   |     |     |   |
| PALA (Ours)                                                                                                           | vs. | GPT-4o mini      | 45%      | 37% |     |     |     |   |
|                                                                                                                       |     | Qwen2-7B         | 13%      | 15% |     |     |     |   |
|                                                                                                                       |     | Llama-3-ELYZA-8B | 94%      | 88% |     |     |     |   |
|                                                                                                                       |     | Llama-3-youko-8b | 88%      | 78% |     |     |     |   |

**Table 3**

Correctness of conclusions with and without multi-agent approach or expert guidance (trading day 0-1).

|                           | MCC          | Macro F1     | Pos F1 | Neg F1 | Diff   |
|---------------------------|--------------|--------------|--------|--------|--------|
| GPT-4o mini: Single-agent |              |              |        |        |        |
| (a) w/o expert guidance   | 0.034        | 0.502        | 0.348  | 0.656  | -0.309 |
| (b) w/ expert guidance    | 0.127        | 0.559        | 0.534  | 0.584  | -0.050 |
| GPT-4o mini: Multi-agent  |              |              |        |        |        |
| (c) w/o expert guidance   | 0.092        | 0.541        | 0.428  | 0.653  | -0.225 |
| (d) w/ expert guidance    | <b>0.156</b> | <b>0.570</b> | 0.557  | 0.583  | -0.026 |
| Qwen2-7B: Single-agent    |              |              |        |        |        |
| (a) w/o expert guidance   | 0.042        | 0.521        | 0.433  | 0.608  | -0.176 |
| (b) w/ expert guidance    | 0.065        | 0.530        | 0.429  | 0.631  | -0.202 |
| Qwen2-7B: Multi-agent     |              |              |        |        |        |
| (c) w/o expert guidance   | 0.093        | 0.533        | 0.536  | 0.530  | 0.006  |
| (d) w/ expert guidance    | <b>0.109</b> | <b>0.549</b> | 0.526  | 0.571  | -0.045 |

**Table 4**

Quality of analysis with and without multi-agent approach or expert guidance.

|                              |     |                               | Criteria | 1    | 2   | 3   | 4   | 5   |
|------------------------------|-----|-------------------------------|----------|------|-----|-----|-----|-----|
| GPT-4o mini                  |     |                               |          |      |     |     |     |     |
| (b) Single-agent w/ guidance |     |                               |          | 91%  | 97% | 56% | 14% | 5%  |
| (c) Multi-agent w/o guidance | vs. | (a) Single-agent w/o guidance |          | 100% | 76% | 99% | 46% | 47% |
| (d) Multi-agent w/ guidance  |     |                               |          | 99%  | 99% | 96% | 88% | 89% |
| Qwen2-7B                     |     |                               |          |      |     |     |     |     |
| (b) Single-agent w/ guidance |     |                               |          | 20%  | 12% | 12% | 7%  | 2%  |
| (c) Multi-agent w/o guidance | vs. | (a) Single-agent w/o guidance |          | 99%  | 76% | 58% | 97% | 83% |
| (d) Multi-agent w/ guidance  |     |                               |          | 92%  | 91% | 73% | 71% | 78% |

are beyond the scope of this paper. Previous studies using textual information for stock price predictions do not always distinguish this point. For example, predicting Day  $t + 1$  stock returns using news up to 23:59 of Day  $t$  but basing stock returns on Day  $t$ 's closing price does not guarantee profitability.

#### 4.6. Ablation study 1: Effect of multi-agent approach and expert guidance

We further investigated the effect of multi-agent approach and expert guidance on analysis generation. We test for four patterns:

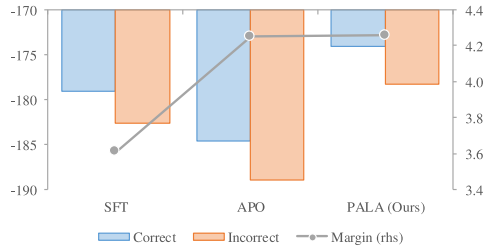
single-agent (a): without expert guidance and (b): with expert guidance, and multi-agent (c): without expert guidance and (d): with expert guidance. (a) corresponds to setting 1 in Section 4.2, (c) corresponds to setting 2 and (d) corresponds to setting 3. In (b), all expert guidance was added to the prompt of (a). Table 3 presents the correctness of conclusions. Compared to pattern (a), either (b) adding expert guidance or (c) applying multi-agent approach can improve the performance. By incorporating both multi-agent and expert guidance, (d) achieves better performance than others. Furthermore, when comparing the differences between F1 scores for positive rating and negative rating samples,

**Table 5**  
Correctness of conclusions across models (trading day 0-1).

|             | MCC          | Macro F1     | Pos F1 | Neg F1 | Diff   |
|-------------|--------------|--------------|--------|--------|--------|
| Qwen2-7B    | 0.125        | 0.534        | 0.568  | 0.500  | 0.068  |
| SFT         | 0.093        | 0.530        | 0.542  | 0.517  | 0.025  |
| APO         | 0.164        | 0.573        | 0.562  | 0.584  | -0.023 |
| PALA (Ours) | <b>0.180</b> | <b>0.583</b> | 0.583  | 0.566  | 0.017  |

**Table 6**  
Quality of analysis across models.

|             |     |          | Criteria |     |
|-------------|-----|----------|----------|-----|
|             |     |          | 4        | 5   |
| SFT         | vs. | Qwen2-7B | 28%      | 11% |
| APO         |     | SFT      | −10%     | 4%  |
| PALA (Ours) |     | APO      | 5%       | 3%  |



**Fig. 3.** Log-likelihood of generating correct and incorrect responses across models.

(d) tends to show reduced disparity compared to the other patterns, indicating not only a higher overall capability but also a more balanced ability to make both positive and negative judgments for fundamental analysis. These results demonstrate the effectiveness of our approach to break down the analysis process into multi-stages and apply an agent with expert guidance in each stage. Additional performance metrics can be found in [Appendix A.3](#).

[Table 4](#) presents the analysis quality. Compared to pattern (a), (b) through (d) all improve the analysis quality across all five criteria. However, the improvement of (b) single-agent with expert guidance, is relatively modest compared to the multi-agent approaches (c) and (d), especially for criteria 4 and 5. This suggests that in a single response, LLMs are not capable of conducting a sufficient analysis, underscoring the importance of our proposed multi-agent approach.

#### 4.7. Ablation study 2: Effect of PALA

[Table 5](#) shows the correctness of conclusions for the base model Qwen2-7B, fine-tuned with SFT, APO and our PALA. SFT trained with only correct analyses did not result in better performance than the base model. After fine-tuning the model with APO, which also incorporates incorrect analyses, performance improved. Our proposed method, PALA, further enhances the performance. Additionally, the difference between the F1 scores for positive rating and negative rating samples was the smallest with PALA, indicating a more balanced ability to distinguish between positive and negative ratings. Additional performance metrics can be found in [Appendix A.3](#).

[Table 6](#) presents the analysis quality. SFT performs better than the base model on both criteria 4 and 5, which assess comprehensiveness and logical soundness. APO underperforms SFT in criterion 4 but outperforms it in criterion 5. We observed that SFT sometimes extracts unimportant key points, leading to a higher score in criterion 4, although this does not contribute to the final conclusion. PALA slightly surpasses APO in both criteria 4 and 5.

We quantitatively assessed the impact of PALA on likelihood changes. [Fig. 3](#) displays the average log-likelihood of generating correct and

**Table 7**  
Correctness of conclusions based on key points identified by PALA.

|             | MCC   | Macro F1 | Pos F1 | Neg F1 |
|-------------|-------|----------|--------|--------|
| GPT-4o mini | 0.204 | 0.569    | 0.605  | 0.532  |
| (original)  | 0.156 | 0.570    | 0.557  | 0.583  |
| Qwen2-7B    | 0.154 | 0.556    | 0.574  | 0.538  |
| (original)  | 0.125 | 0.534    | 0.568  | 0.500  |

**Table 8**  
Correctness of conclusions based on gold key points.

|             | MCC   | Macro F1 | Pos F1 | Neg F1 |
|-------------|-------|----------|--------|--------|
| GPT-4o mini | 0.452 | 0.700    | 0.712  | 0.688  |
| Qwen2-7B    | 0.328 | 0.632    | 0.657  | 0.606  |
| PALA        | 0.794 | 0.895    | 0.884  | 0.907  |

incorrect analysis  $\log \pi(y_{te}|x)$  and  $\log \pi(y_l|x)$ , respectively. These log-likelihood were calculated for each model when prompted with the analysis  $x$  generated by the Business Analyzer in the test dataset. Compared to SFT, APO results in a greater margin between correct and incorrect analyses while showing a decline in the log-likelihood of both. PALA increases the log-likelihood of correct analyses while maintaining the same margin as APO. This confirms that PALA achieves the intended effects.

#### 4.8. Analysis: Ability enhancement by fine-tuning

The experiments above confirmed that fine-tuning enhances the ability to derive correct conclusions. This ability in our study consists of two components: identifying key points and deducing ratings. This section examines improvements in both areas.

Firstly, to verify whether the fine-tuned model correctly identifies key points, we conducted an experiment using key points identified by the PALA fine-tuned model to have Qwen2 and GPT-4o deduce ratings. As shown in [Table 7](#), using key points from the fine-tuned model, Qwen2 demonstrated better performance in both MCC and F1 compared to using its originally identified key points. For GPT-4o, MCC improved while F1 scores remained similar. This confirms that fine-tuning enhanced the model ability to identify key points.

Next, to assess the ability to deduce ratings, we created correct analyses using the method outlined in [Section 3.2](#). Based on key points in these analyses, we use Qwen2, GPT-4o, and the fine-tuned model to deduce ratings. As illustrated in [Table 8](#), the fine-tuned model achieved superior performance across all metrics.

These experiments demonstrate that fine-tuning effectively enhances both the ability to identify key points and to deduce ratings.

## 5. Conclusion

In this paper, to achieve in-depth fundamental analysis, we proposed an “Analytical Stage-Based Multi-Agent Framework”. This framework emulates the analytical process of human fundamental analysts by breaking down the analysis process into multiple stages, where each stage builds upon the conclusions of the previous one. We assign an LLM agent to each stage, and enhance the capability of each agent with prompting or fine-tuning. Prompting is used to enable agents to efficiently acquire more specialized analysis methods, while fine-tuning improves agent’s ability to extract key information from diverse data and derive conclusions. Together, in sequence, these agents with enhanced capability emulate the complete analytical workflow of a human analyst. Furthermore, we proposed an automatic way to construct the training data and a new method for fine-tuning LLMs. The training data is created by having LLMs generate analyses with specified conclusions. The method for fine-tuning is called Preference-Anchored Likelihood Adjustment. After fine-tuning LLMs to align its

**Financial Analyzer:**

1. In case of full year earnings:
  - 1.1 Determine whether there is any change in the total dividends for the current period since the last disclosure.
  - 1.2 If the company guidance for the total dividends for the next period is provided, state whether it will increase or decrease compared to the current period.
  - 1.3 Calculate the difference between actual full year earnings and the company guidance released previously.
2. In case of quarter earnings:
  - 2.1 Calculate the change rate of the full-year company guidance from the last disclosed guidance.
  - 2.2 Determine whether there is any change in the dividends for the current full year since the last disclosure.
  - 2.3 Calculate the progress rate and compare it with the same period of the previous year.

(a) Expert guidance for Financial Analyzer

**Business Analyzer:**

1. When analyzing the changes in earnings, consider the qualitative circumstances behind it.
2. When analyzing company guidance, analyze the drivers and also consider future developments.
3. Perform an analysis of the shareholder return policy as well.

(b) Expert guidance for Business Analyzer

**Fig. 4.** Expert guidance for Financial Analyzer and Business Analyzer.**Financial Analyzer:**

You are a financial analyst.  
 To analyze the {fiscal\_period} earnings of {company\_name}, please use the following documents:  
 {financial data}  
 #  
 Please analyze the financial data.  
 Use the following instructions as a reference for your analysis:  
 {Expert guidance for Financial Analyzer}

(a) Prompt for Financial Analyzer

**Business Analyzer:**

You are a financial analyst.  
 To decide the rating for {company\_name}, you are to perform a comprehensive analysis on the performance of {company\_name}. First, create an outline of the factors to consider. Refer to the following documents:  
 {qualitative information}  
 #  
 {analysis by Financial Analyzer}  
 #  
 Based on the outline, analyze the performance of {company\_name}. Use the following instructions as a reference for your analysis:  
 {Expert guidance for Business Analyzer}

(b) Prompt for Business Analyzer

**Conclusion Deriver:**

You are a financial analyst.  
 Below is the analysis of the {fiscal\_period} earnings of {company\_name}:  
 {analysis by Business Analyzer}  
 From the above analysis, extract positive and negative factors, and assign a rating to {company\_name}. Select from one of the following ratings:  
 - Positive: Performance is expected to improve.  
 - Negative: Performance is expected to decline.

(c) Prompt for Conclusion Deriver

**Fig. 5.** Prompts for each agent. {} are placeholders.

preference, we introduce a novel objective function to increase the likelihood of generating correct responses while preserving the fine-tuned preference. Experiments on Japanese listed companies show that: 1. Our proposed multi-agent framework can enhance LLMs' correctness in conclusions even without expert guidance and fine-tuning, as it enables more comprehensive and detailed analysis; 2. The framework's analytical performance is further improved when incorporating expert guidance and existing fine-tuning methods; 3. Furthermore, our PALA fine-tuning leads to even better performance.

**CRedit authorship contribution statement**

**Tao Xu:** Conceptualization, Formal analysis, Methodology, Software, Validation, Investigation, Writing – original draft. **Zhe Piao:** Conceptualization, Investigation, Data curation, Funding acquisition. **Tadashi Mukai:** Conceptualization, Investigation. **Yuri Murayama:** Conceptualization, Writing – review & editing. **Kiyoshi Izumi:** Conceptualization, Writing – review & editing, Funding acquisition, Supervision.

```

### 1. Evaluation of Financial Performance (omitted...)
### 2. Analysis of Financial Performance (omitted...)
### 3. Analysis of Performance Outlook (omitted...)
### 4. Analysis by Business Segment (omitted...)
### 5. External Environmental Factors (omitted...)
### 6. New Medium-Term Management Plan (omitted...)
### 7. Positive Factors
1. Increase in Sales: Sales increased by 29.9% year-on-year, with continued strong performance in the logistics business.
2. Increase in Operating Profit: Operating profit increased by 38.5% year-on-year, driven by the strong performance of the logistics business.
3. Significant Increase in Ordinary Profit: Ordinary profit increased by 92.7% year-on-year, significantly impacted by the yen's depreciation.
4. Increase in Net Profit: Net profit increased by 71.7% year-on-year, mainly due to the increase in ordinary profit.
5. Dividend Increase: Dividends increased by 50 yen from the previous forecast, showing an enhanced commitment to shareholder returns.
6. Upward Revision of Full-Year Guidance: The full-year guidance for sales, operating profit, ordinary profit, and net profit have been revised upwards from the previous forecast.
7. Promotion of ESG Management: Environmental investments and efforts toward decarbonization are being advanced, indicating potential for long-term growth.
8. Strengthening of Capital Policy: Measures such as the introduction of DOE, share buybacks, and a review of the dividend payout ratio are being considered, indicating strengthened shareholder returns.
### 8. Negative Factors
1. Risk of Freight Rate Decline: There is a risk of declining freight rates in the liner shipping and logistics businesses.
2. Global Economic Uncertainty: Inflation and economic slowdown could impact performance.
3. Exchange Rate Risk: While the yen's depreciation has contributed to performance, future exchange rate risks should be noted.
4. Fluctuation in Fuel Prices: Variability in fuel oil prices could impact performance.
5. Regulatory Risk: Strengthening of environmental regulations could impact performance.
### 9. Rating
There are many positive factors, particularly the increase in sales and profits, dividend increase, and promotion of ESG management, which are highly valued. Although there are negative factors such as the risk of declining freight rates, global economic uncertainty, and exchange rate risk, the positive factors currently outweigh them.
Given that performance is expected to improve in the future, the rating is "Positive."

```

Fig. 6. An example of analysis by our multi-agent framework.

| Financial data          |                       | Qualitative information |
|-------------------------|-----------------------|-------------------------|
| Current quarter results | Fiscal year guidance  |                         |
| Revenue, YoY            | Revenue, YoY          | Financial conditions    |
| Operating Profit, YoY   | Operating Profit, YoY | Business conditions     |
| Ordinary Profit, YoY    | Ordinary Profit, YoY  | Earnings guidance       |
| Net Profit, YoY         | Net Profit, YoY       |                         |
| Shareholder return      | Shareholder return    |                         |

Fig. 7. Information from earnings reports used in our experiments.

**For Financial Analyzer:**

- Use the financial data of the current fiscal period, the same period of the previous year, and the previous quarter.

**For Business Analyzer:**

- Use the processed financial data for the current period and qualitative information of the current period.

**For Conclusion Deriver:**

- Use the comprehensive analysis for the current period.

Fig. 8. Rules of information selection for each agent.

**Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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**Appendix****A.1. Prompts and other supplementary information**

\* All prompts and the analysis example are translated from Japanese to English.

**A.2. Analysis of the PALA objective function**

First, we derive the gradient of PALA objective:

$$\nabla_{\theta'} \mathcal{L}_{PALA}(x, y_w, y_l; \theta') = -\nabla_{\theta'} \sigma(r_{\theta'}(x, y_w)) - \nabla_{\theta'} \text{Anc}(rm_{\theta'}(x, y)) \quad (11)$$

**Table 9**

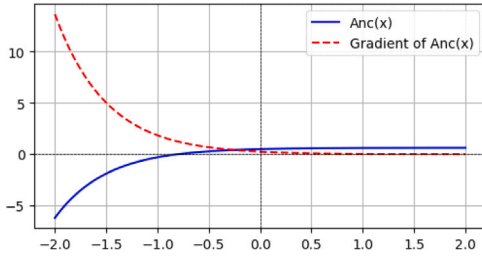
Performance comparison with and without multi-agent approach or expert guidance (trading day 0-1).

|                           | Precision    |              |              | Recall       |              |              |
|---------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                           | Macro        | Pos          | Neg          | Macro        | Pos          | Neg          |
| GPT-4o mini: Single-agent |              |              |              |              |              |              |
| (a) w/o expert guidance   | 0.519        | 0.453        | 0.585        | 0.515        | 0.282        | <b>0.748</b> |
| (b) w/ expert guidance    | 0.563        | 0.486        | 0.640        | 0.564        | 0.592        | 0.536        |
| GPT-4o mini: Multi-agent  |              |              |              |              |              |              |
| (c) w/o expert guidance   | 0.548        | 0.490        | 0.607        | 0.544        | 0.380        | 0.707        |
| (d) w/ expert guidance    | <b>0.577</b> | <b>0.495</b> | <b>0.659</b> | <b>0.579</b> | <b>0.635</b> | 0.522        |
| Qwen2-7B: Single-agent    |              |              |              |              |              |              |
| (a) w/o expert guidance   | 0.521        | 0.451        | 0.592        | 0.521        | 0.416        | 0.626        |
| (b) w/ expert guidance    | 0.533        | 0.468        | 0.599        | 0.531        | 0.396        | <b>0.667</b> |
| Qwen2-7B: Multi-agent     |              |              |              |              |              |              |
| (c) w/o expert guidance   | 0.547        | 0.464        | 0.629        | 0.547        | <b>0.635</b> | 0.458        |
| (d) w/ expert guidance    | <b>0.554</b> | <b>0.476</b> | <b>0.632</b> | <b>0.555</b> | 0.588        | 0.522        |

**Table 10**

Performance comparison across models (trading day 0-1).

|             | Precision    |              |              | Recall       |              |              |
|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             | Macro        | Pos          | Neg          | Macro        | Pos          | Neg          |
| Qwen2-7B    | 0.565        | 0.470        | 0.659        | 0.560        | <b>0.718</b> | 0.403        |
| SFT         | 0.547        | 0.463        | 0.632        | 0.546        | 0.655        | 0.438        |
| APO         | 0.581        | 0.498        | 0.664        | 0.582        | 0.643        | 0.522        |
| PALA (Ours) | <b>0.589</b> | <b>0.508</b> | <b>0.670</b> | <b>0.591</b> | 0.639        | <b>0.542</b> |

**Fig. 9.**  $\text{Anc}(x)$  and its gradient.

Here, we define

$$\delta(x) := \partial\sigma(x)/\partial x = \sigma(x)(1 - \sigma(x))$$

and note that

$$rm_{\theta'}(x, y) = r_{\theta'}(x, y_w) - r_{\theta'}(x, y_l) = \beta \log \frac{\pi_{pala}(y_w|x)}{\pi_{apo}(y_w|x)} - \beta \log \frac{\pi_{pala}(y_l|x)}{\pi_{apo}(y_l|x)} \quad (12)$$

We can obtain the gradient for each case of  $rm_{\theta'}(x, y) \geq 0$  and  $rm_{\theta'}(x, y) < 0$ :

(1) When  $rm_{\theta'}(x, y) \geq 0$ :

$$\begin{aligned} \nabla_{\theta'} \mathcal{L}_{PALA}(x, y_w, y_l; \theta') = & \\ & -\beta \underbrace{\delta(r_{\theta'}(x, y_w))}_{\text{higher weight when winning reward close to 0}} \underbrace{\nabla_{\theta'}(\log \pi_{pala}(y_w|x))}_{\text{increase winning likelihood}} \\ & -\beta \underbrace{\delta(rm_{\theta'}(x, y))}_{\text{higher weight when reward margin close to 0}} \underbrace{\nabla_{\theta'}(\log \pi_{pala}(y_w|x) - \log \pi_{pala}(y_l|x))}_{\text{increase likelihood difference}} \end{aligned} \quad (13)$$

(2) When  $rm_{\theta'}(x, y) < 0$ :

$$\begin{aligned} \nabla_{\theta'} \mathcal{L}_{PALA}(x, y_w, y_l; \theta') = & \\ & -\beta \underbrace{\delta(r_{\theta'}(x, y_w))}_{\text{higher weight when winning reward close to 0}} \underbrace{\nabla_{\theta'}(\log \pi_{pala}(y_w|x))}_{\text{increase winning likelihood}} \end{aligned}$$

$$-\beta \underbrace{ak \exp(k(rm_{\theta'}(x, y)))}_{\text{sharply increase weight when reward margin below 0}} \underbrace{\nabla_{\theta'}(\log \pi_{pala}(y_w|x) - \log \pi_{pala}(y_l|x))}_{\text{increase likelihood difference}} \quad (14)$$

The objective function aims to increase the winning likelihood and the likelihood difference between winning and losing responses. If the reward margin becomes negative during training, a high magnitude is applied to increase the likelihood difference, thus pushing the reward margin back to positive.

Fig. 9 illustrates the shape of  $\text{Anc}(x)$  and its gradient. The coefficients are set as  $a = -0.125$ ,  $k = -2$ , and  $b = 0.625$ , aligning with the experiment in Section 4. As  $x$  moves below 0,  $\text{Anc}(x)$  decreases exponentially and its gradient increases exponentially.

Next, we check the reward of winning and losing responses and their margins during training with APO and PALA in the experiment of Section 4. As shown in Fig. 10(a), during training with APO, while the reward margin expands, winning rewards decreases, meaning the likelihood of  $\pi_{apo}$  to generate winning responses declines against the SFT model  $\pi_{sft}$ . Further training  $\pi_{apo}$  with PALA, the winning rewards increases, and the reward margin are kept around zero against  $\pi_{apo}$  as shown in Fig. 10(b), meaning the preference learned through APO is preserved and the likelihood to generate winning responses increased. We can confirm the effect of PALA directly against SFT in Fig. 10(c). We also did experiments to exclude  $\text{Anc}(x)$  in the PALA objective, and the reward margin decreases during PALA training, demonstrating the necessity of the term  $\text{Anc}(x)$ .

### A.3. Additional experimental results

Tables 9–10 present the precision and recall metrics across different settings and models. The multi-agent approach with expert guidance achieves the highest scores in macro precision and recall, as well as in precision for both positive and negative ratings. PALA exhibits the similar results.

### Data availability

The authors do not have permission to share data.

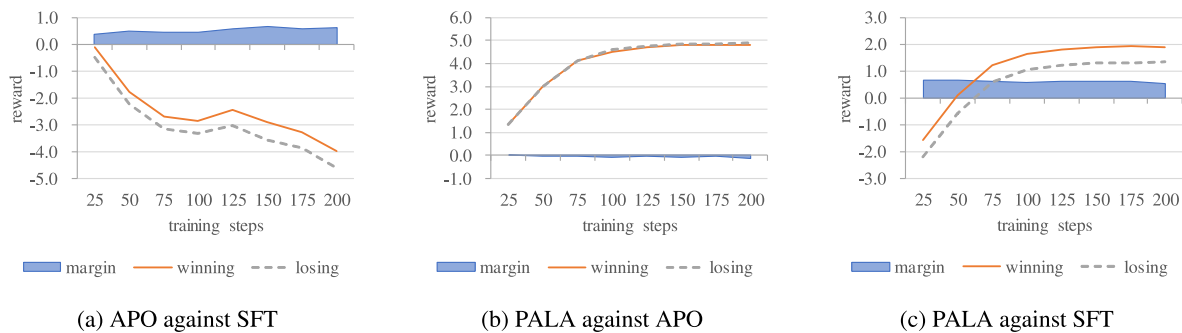


Fig. 10. Reward of winning and losing responses in the validation dataset during training in experiments of Section 4.

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